



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0510 – COMPUTER VISION

EXPERIMENT NO: 17

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17. Perform Edge detection using Sobel Matrix along X axis

AIM:

Edge detection using Sobel Matrix along X axis

PROGRAM:

```
import cv2

img = cv2.imread(r"C:\Users\premarubha\Desktop\cv model image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

sobelx = cv2.convertScaleAbs(sobelx)

cv2.imshow("Original Image", img)

cv2.imshow("Sobel X Edge Detection", sobelx)

cv2.imwrite(r"C:\Users\premarubha\Desktop\sobel_x_image.jpg", sobelx)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.